

Johann Schmidt

Research Scientist in Deep Learning

he/him · born 1995

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Applied research scientist in deep learning with six years of full-time R&D alongside industry partners. I take problems from framing to a working system, from production scheduling at an electronics manufacturer to traffic signal control and acoustic sensing on edge hardware. My PhD makes frozen pretrained vision models robust to new perspectives of the input signal without retraining. This is key for safe and reliable inference of foundation models. I was named a Top Reviewer at NeurIPS 2026 and have publications at top-tier venues like ICML. Open to senior applied scientist roles from early 2027, with the long-term goal of leading applied research in industry.

6+	18+	15+	12+
Years Work Experience	Theses Supervised	Papers Reviewed	Papers Published

EXPERIENCE

Research Scientist at OVGU · AILab, Otto von Guericke University Magdeburg, Germany 2020 – 2027
Full-time position on funded R&D projects with industry partners, with the PhD pursued alongside.

- **KIVA-NET** · with Thorsis Technologies GmbH 2025 – 2027
  Low-cost acoustic sensor that counts and classifies vehicles from sound alone, in real time on edge hardware. Analytical signal processing pre-filters the audio, so only vehicle passes reach a frozen audio foundation model with a linear probe. 96.5% detection accuracy on more than 16,000 auto-labeled passes. Components from PASCAL and KIVA-NET are the basis of a multi-modal traffic sensor that Thorsis is developing.
- **PASCAL** · with Thorsis Technologies GmbH 2022 – 2025
  Traffic signal control on real-time V2X data for Magdeburg's urban testbed. Developed TransferLight with a Master's student I supervised, one reinforcement learning policy on a hierarchical graph neural network for any intersection layout. Trained only on synthetic networks, it left 54% fewer vehicles standing than fixed-time control at an unseen real intersection. Deployed the demonstrator as FastAPI services on two NVIDIA Jetson AGX Orin edge nodes, first rolled out in a SUMO simulation I largely wrote. Thorsis took it into practice after the project.
- **AI-Engineering** · joint project of five universities in Saxony-Anhalt 2022
  New Bachelor's program in AI and engineering by five universities, running since 2023. Contributed to the curriculum and led the development of OERport, the cross-university course platform (Next.js, Shibboleth SSO, MySQL).
- **SENECA** · with Thorsis Technologies GmbH and TECTRON GmbH 2020 – 2022
  Real-time production scheduling for the electronics manufacturer Tectron, NP-hard with up to 200 open orders per run. Methods published at ESWA, IFAC INCOM and HICSS. Delivered a planning tool with heuristic and deep learning methods to choose from, which beat manual planning on late orders and total tardiness in production. On 170 real orders, our simulated annealing method had no late orders and 23% fewer setups than a genetic algorithm, in 0.7 s instead of 66 s.

EDUCATION

PhD in Deep Learning at OVGU · Otto von Guericke University Magdeburg, Germany 2022 – 2026
Thesis research since 2022, alongside the full-time project work. Submission at the end of 2026; the PhD ends in April 2027.

- **PhD Thesis** · AILab, Otto von Guericke University Magdeburg, Germany
  Robust and Efficient Discriminative Deep Learning by Canonicalizing and Lifting Symmetry-Degenerated Signals. A small module in front of a frozen pretrained model corrects the view of each input to ease the downstream task, which adds robustness across views. The methods act as preprocessing add-ons to frozen off-the-shelf models.
- **OxML Summer School** · Oxford, United Kingdom August 2025

Sustainable Supply Chain DeepHack · 2nd place, Transatlantic AI Hackathon 2022
Built in one weekend with a team: automatic parcel volume estimation with joint planning of vehicle loading and route.

Master: Digital Engineering · Otto von Guericke University Magdeburg, Germany 2018 – 2020
Grade 1.4. Thesis at Fraunhofer IFF on vision-based human action recognition for human-machine interaction in industry, grade 1.1, published at ETFA 2021.

Bachelor: Electrical Engineering · Magdeburg-Stendal University of Applied Sciences, Germany 2014 – 2018
Grade 1.7. Specialization in process automation. Thesis at ifak Magdeburg on a GUI for an ultrasonic sensor system on an embedded STM32 controller in C, grade 1.2.

SELECTED PUBLICATIONS

ECML-PKDD 2026
Zero-Shot Test-Time Canonicalization using Out-of-Distribution Scoring
D. Lindner, J. Schmidt, T. Siegl, M. Becker, S. Stober. Joint European Conference on Machine Learning and Knowledge Discovery in Databases.

BMVC 2026
PPS: Plug-and-Play Saccadic Vision for Fine-Grained Classification
J. Schmidt, S. Stober, J. Denzler, P. Bodesheim. British Machine Vision Conference.

ECCV: BISCUIT 2026
Geometrically Constrained and Token-Based Probabilistic Spatial Transformers
J. Schmidt, T. Siegl, M. Becker, S. Stober. European Conference on Computer Vision: International Workshop on Biomedical Image and Signal Computing for Unbiasedness, Interpretability, and Trustworthiness.

AAAI: MALTA 2025
TransferLight: Zero-Shot Traffic Signal Control on any Road-Network
J. Schmidt, F. Dreyer, S. A. Hashimi, S. Stober. Association for the Advancement of Artificial Intelligence: Multi-Agent Reinforcement Learning for Transportation Autonomy Workshop.

ICML 2024
Tilt your Head: Activating the Hidden Spatial-Invariance of Classifiers
J. Schmidt, S. Stober. International Conference on Machine Learning.

HICSS 2024
Reviving Simulated Annealing: Lifting its Degeneracies for Real-Time Job Scheduling
J. Schmidt, B. Köhler, H. Borstell. Hawaii International Conference on System Sciences.

ESWA 2021
NeuroEvolution of augmenting topologies for solving a two-stage hybrid flow shop scheduling problem: A comparison of different solution strategies
S. Lang, T. Reggelin, J. Schmidt, M. Müller, A. Nahhas. Expert Systems with Applications.

SKILLS

Tasks and Data	Time series, segmentation, classification, regression, ranking, point clouds, 2D/3D object detection, audio events, graphs, image generation, gaze tracking, quantum control, NeRFs
LLM Tooling	llama.cpp for local models, Claude Code with MCP, agents and skills
Deep Learning Stack	PyTorch (Lightning, Geometric), Hugging Face, timm, OpenCV, Kornia, escnn; MLOps with W&B, MLflow, Optuna, Hydra; scikit-learn; earlier TensorFlow
Software Engineering and Backend	Git, CI pipelines, clean code, DevOps; FastAPI, Flask; PostgreSQL, MySQL; agile workflow with Scrum, Jira, Confluence, Trello
Simulation and Optimization	Discrete event simulation (logistics), SUMO (traffic), Google OR-Tools (combinatorial optimization)
Multi-GPU Training	Multi-node, multi-zone heterogeneous Ubuntu GPU cluster with distributed training (Hugging Face Accelerate); TensorRT and edge deployment
Operating Systems	Linux (Arch, Ubuntu, Mint, Kali), Windows, macOS
Programming Languages	Python (main language); C and Assembler (microcontrollers), C++; Java (Android apps); C# (Unity games); JavaScript, HTML5, CSS, Tailwind CSS, Jekyll (web)
Spoken Languages	German (native), English (daily working language)
Mobility	German driving licence; willing to travel regularly, including international trips

LEADERSHIP AND SERVICE

Reviewing	15 papers for ICLR 2027, NeurIPS 2026, ECCV 2026, ECCV: BISCUIT 2026, NLDL 2026. Recognized as **Top Reviewer** at NeurIPS 2026.
Paper Club	Initiated and co-moderated a biweekly paper club at OVGU.
Thesis Supervision	18 Bachelor's and Master's theses, many with partners such as Fraunhofer IFF, Trumpf and SaxonQ. They led to two publications (AAAI workshop 2025, ECML-PKDD 2026).

TALKS AND TEACHING

Conference Oral	ECML-PKDD (Main Research Track and QCDS Workshop), 2026; AAAI (MALTA Workshop), 2025; HICSS, 2024; IFAC INCOM, 2021
Invited Talk	Friedrich Schiller University Jena, 2025
Seminar	Deep Learning Project Seminar on traffic signal control with reinforcement learning and graph neural networks, designed and taught on my own (winter semester 2024/25)
Training	Science communication seminar, NaWik (2023); eleven teaching workshops, OVGU (2021)